

CHIMERA-QRC: A Pre-Registered, Adversarially-Controlled Study of Quantum Reservoir Computing for S&P 500 Realized-Volatility Forecasting

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1. Focus, track selection, and problem framing

We target **Track A: financial volatility** — one-step-ahead forecasting of S&P 500 **realized variance (RV) and the tracking of volatility** regime transitions, the domain where JonesTrading's risk, allocation and derivatives pricing live. RV is long-memory, multi-scale, regime-switching and non-Gaussian — the structure reservoir computing exploits. Our central yardsticks are the Echo State Network (ESN, the classical analog of QRC) and the strongest econometric models (HAR — *Heterogeneous AutoRegressive*, Corsi 2009 — and, critically for Phase 3, **HAR-X**: HAR augmented with the same realized-measure features we feed the quantum reservoir). We also report GARCH/GJR-GARCH, AR(3), persistence, LSTM, and an RFF/RBF kernel, and we implement the challenge's common **MNIST** benchmark on the identical engine.

This paper's honest thesis. We pre-registered a falsifiable advantage hypothesis (H0) and then *attacked our own result* with the strongest fair controls. The finding is a **carefully bounded negative: at simulable scale (≤ 16 qubits) the quantum reservoir shows no statistically significant forecasting advantage** over strong classical baselines once they are given the same information; the dominant lever is *which features are encoded*, not the reservoir's quantum nonlinearity. We report what *does* survive — measurable kernel distinctness, lower seed variance, and full-rank entanglement (classical-simulation hardness) — as quantum-specific *properties* that are necessary, not sufficient, for advantage.

Vs. prior work. The closest study (Li et al. 2025/26 — QRC for realized volatility) presents the approach as a competitive, noise-resilient *proof-of-concept*; we add the control-hardened, pre-registered evaluation it omits — the decisive **HAR-X** control, HAC-DM/Holm/MCS, and explicit capability and per-layer-noise audits — turning an encouraging proof-of-concept into a falsifiable result.

2. QRC architecture (Fig. 1)

CHIMERA is a delay-embedding quantum reservoir. Inputs are angle-encoded $R_Y(\pi \cdot x)$, one value per qubit, onto $|0\dots 0\rangle$; the state evolves under a fixed transverse-field **Ising** Hamiltonian $H = \sum_{\{i<j\}} J_{ij} Z_i Z_j + h_x \sum_i X_i$ ($h_x=1$; J a **random $\approx 50\%$ -connected** graph, $\text{connectivity}=0.5$; $U=\exp(-iH\tau)$, $\tau=2$); single- and pairwise Pauli-Z expectations $\langle Z_i \rangle$, $\langle Z_i Z_j \rangle$ form the feature vector. A **ridge** head consumes these features concatenated with a linear block of the same inputs (incl. HAR), so any reservoir gain is genuine nonlinearity *beyond* the linear span of identical information. The identical inputs feed the classical controls, isolating quantum-vs-classical. Four mechanisms extend the core (multi-scale τ -bank; RZ measurement feedback; regime-adaptive Ising $\leftrightarrow$ Heisenberg via BOCPD; dissipation-as-feature); Phase 3 adds an **encoding-density / data-re-uploading path** (§5.2) and a **sparse/tensor backend** (§5.5). The engine is pure NumPy; an explicit PennyLane circuit reproduces it to $\approx 5 \times 10^{-4}$ and compiles, at $n=8$, to **380 native gates** (220 two-qubit + 160 one-qubit, 20 Trotter layers) — consistent with the $\approx 50\%$ -connected graph (an all-to-all reservoir would need ≈ 720).

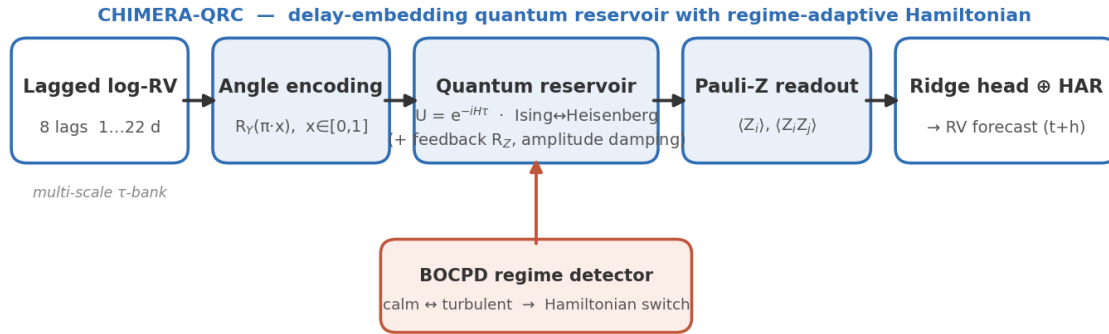


Figure 1. CHIMERA-QRC pipeline: lagged log-RV (and, in Axis B, realized-measure) inputs are angle-encoded onto qubits, evolved under a fixed random-coupled Ising Hamiltonian, and read out as single/pairwise Pauli-Z expectations into a ridge head fused with a linear block of the same inputs; a BOCPD detector selects the Hamiltonian for regime adaptivity.

3. Theoretical and analytical justification

The exploited quantum resources map to RV's structure: (a) **Hilbert-space dimensionality** —

n qubits give 2^n amplitudes and $O(n^2)$ measured Pauli features at $O(n^2)$ parameter cost;

(b) **intrinsic nonlinearity** from fixed unitary evolution (no gradient training, no barren plateaus); (c) **fading memory** (echo-state property); (d) **Hamiltonian as inductive bias**.

The central, falsifiable mechanism is **expressivity scaling**, which we measure directly via the geometric difference (Huang et al. 2021). The quantum kernel is poorly reproducible by a matched classical reservoir: against the name-matched ESN-108 reference,

$g(\text{ESN} \rightarrow \text{CHIMERA}) \approx 62$ vs ≈ 4 control (`cli.py run kernel`); against per- n *feature-matched* ESNs the gap is larger still ($g \approx 125$ at $n=8$, rising to ≈ 158 at $n=10$). (g is at fixed ridge regularization; the qualitative $\sim 15\text{--}40\times$ separation, not the exact value, is the claim.)

Crucially, at $\leq 12\text{--}16$ qubits the map remains classically

simulable, so distinctness is **necessary, not sufficient** for accuracy — and §5.3 shows that, here, it is indeed *not* sufficient.

4. Data modeling strategy

Datasets (all public). Oxford-Man .SPX 5-minute realized variance, 2000–2020, 5,029 supervised days incl. the 2008 GFC (Heber et al. 2009; via R packages `highfrequency/bvhar`); public SPY daily OHLCV 2022–2026 (Garman–Klass proxy); real MNIST (Keras .npz mirror).

Preprocessing: log-RV target; features min-max scaled to $[0,1]$ on **train only**; per-model ridge penalty selected on a chronological validation tail; multi-seed ensembling. We verified the pipeline is leakage-free (all features lagged ≥ 1 day; train-only scaling). **Baselines:** HAR, **HAR-X**, GARCH(1,1), GJR-GARCH, AR(3), persistence, ESN (matched + $4\times$, recurrent), RFF kernel, LSTM. **Metrics:** RMSE(log-RV), QLIKE (Patton 2011), Mincer–Zarnowitz R^2 ; significance by **Diebold–Mariano with a Newey–West HAC variance** (HLN-corrected), **Holm**-adjusted across the comparison family, and the Model Confidence Set (Hansen–Lunde–Nason 2011).

5. Phase-3 execution — concrete results

Pre-registration (honesty). Before running, we fixed the confirm/refute thresholds in `preregistration.py` (H0/H1/H4, from our Phase-2 §7). All outcomes — including negatives — are reported against them. Full numbers/wall-clock: `results/*_findings.md`.

5.1 The input-bottleneck mechanism (pre-registered negative). With a *fixed* univariate 8-lag encoder, extra qubits receive no new input: $g(n)$ and effective feature-rank

saturate/decline (n=8→12: g 133→30, D_eff 1.8→1.5). **H0 is refuted in this regime** — the empirical case *for* enriching the encoding.

5.2 Encoding density (Axis B) — mechanism confirmed. Feeding the extra qubits genuinely new realized-measure information (signed return/leverage, downside-semivariance share, jump share) restores the lost structure: at n=10, informed vs idle qubits, kernel distinctness and rank jump (g 52→158, D_eff 1.5→3.1), and effective rank grows with n (D_eff 1.81→3.10→3.14, n=8→12; near-flat 10→12). So the bottleneck is *fixable* — added qubits can carry new information. The decisive question is whether this converts into forecasting **accuracy**.

5.3 The decisive, adversarially-controlled test (the honest headline). We compare CHIMERA to the strongest fair baselines — **HAR-X** (the same rich features used *linearly*, no reservoir), a **true recurrent ESN**, and an **RFF kernel** — all sharing the identical information set and *nesting* HAR-X, so a quantum win requires nonlinearity beyond the linear span. With **8 seeds**, **HAC-DM**, **two windows**, and **Holm correction** (`cli.py run axisB_rig`):

RMSE(log-RV)	HAR-X	recurrent ESN	RFF	CHIMERA	best
crisis n=8	0.6255	0.6386	0.6302	0.6379	HAR-X
crisis n=10	0.6034	0.5996	0.6047	0.6074	ESN
crisis n=12	0.5906	0.6023	0.5964	0.6031	HAR-X
calm n=10	0.6244	0.6445	0.6264	0.6291	HAR-X

(plain HAR for reference: crisis 0.6290, calm 0.6454 — HAR-X's rich features cut RMSE sharply.)

HAR-X is best or co-best everywhere; CHIMERA never beats it (slightly worse; raw-significant at n=8/12) and is statistically indistinguishable from the classical ESN/RFF. ****After Holm correction no comparison is significant in either direction.**** The earlier "beats HAR" result was an artifact of comparing against a *feature-poor* HAR: the gain comes from the encoded realized measures (a known SHAR/HARQ effect), not from quantum nonlinearity. ****By our pre-registered criteria, H0 is refuted — we report this negative.**** What honestly survives for the quantum reservoir: it is **competitive** (within ~0.5–1.5% RMSE of the best at every n), **more stable than the recurrent ESN** (lower per-seed variance, e.g. n=12: 0.009 vs 0.015), and it **beats the recurrent ESN on the calm window** (raw p=0.018; n.s. after Holm).

Named canonical baselines (`cli.py run canonical`): **SHAR**, **HAR-CJ**, **HARQ** (RQ≈RV² proxy) and **HEAVY-RM**, under MSE- *and* QLIKE-loss DM — **none beats HAR-X** (a fair, strong stand-in, not a strawman), and CHIMERA ties the best on RMSE with only a *raw, non-Holm* QLIKE/MZ edge (`results/canonical_baselines_findings.md`).

Rigorous Axis-B: HAR-X (rich linear) is best; quantum shows no significant edge (Holm n.s.)

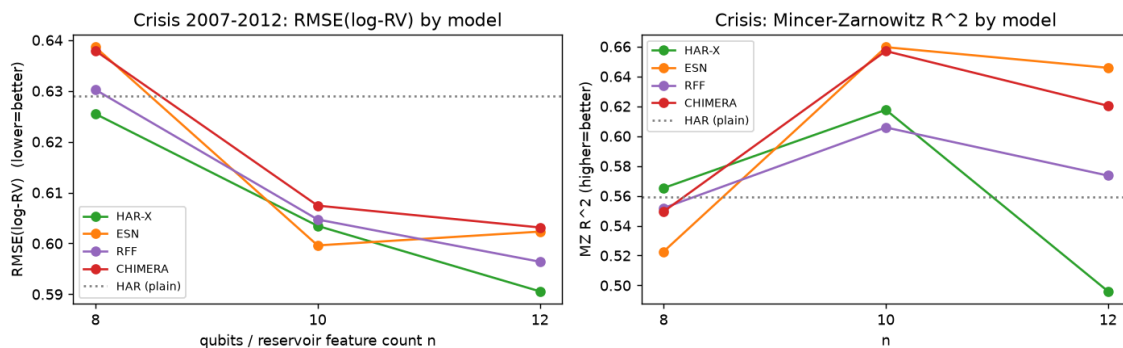


Figure 2. Rigorous Axis-B (8 seeds, crisis window). HAR-X (rich features, linear) is the best or co-best model at every n; the quantum reservoir is competitive but shows no advantage that survives HAC-DM + Holm correction. Left: RMSE(log-RV). Right: Mincer–Zarnowitz R².

5.4 Common MNIST benchmark + noise. Same engine, pixels→PCA(n)→n qubits→Pauli readout→ridge. Accuracy **scales with qubits** (n=5/8/10/12: 0.632/0.798/0.831/0.859, 3 seeds) and ****beats** the linear-PCA baseline at every n (**real nonlinear lift**), **confirming** sufficient expressivity** (the benchmark's purpose); a matched ESN **ties or slightly exceeds** CHIMERA (within ≈1% for n≥8; 2.9% at n=5) — competitive, not dominant. **Noise:** the classifier is ****invariant** to depolarizing** noise — provably, because depolarizing is a uniform Bloch contraction that per-feature standardization removes exactly (accuracy identical across rates 0.05–0.30) — and **robust to amplitude damping** (<0.5% at 30%). **Honesty check** (`cli.py run noise_circuit`): a per-Trotter-layer density-matrix study shows the ***converse*** — two-qubit noise ***during*** evolution is **not** removed by standardization (standardized error grows with rate, vs ≈0 for readout-only depolarizing): the invariance above is a ***readout*** property, not a circuit-level robustness claim; accumulated 2-qubit error over the 380-gate circuit is the real NISQ cost (§6).

5.5 Scaling frontier + quantum-complexity metric. A sparse-exact backend (`expm_multiply`, no dense propagator; matches the dense engine to 2.4×10^{11}) reaches **n=16 exactly**. For the random ≈50%-connected reservoir we measure the **bond dimension** χ_{eff} across a balanced cut: it is **full at every n**, $\chi_{\text{eff}} = 2^{(n/2)}$ (16→256 for n=8→16) — an exact MPS gets ****zero compression**** — with entanglement entropy $S \approx 1.7\text{--}3.2$ nats. The reservoir therefore admits **no low-bond-dimension (MPS/TEBD) shortcut** — exact cost stays exponential (full entanglement is ***necessary**, not sufficient* for true classical hardness) — the precondition any beyond-frontier advantage would need, even though no advantage appears at the simulable scale we can test. **Capability + efficiency checks.** An information-processing-capacity probe (Dambre et al. 2012; `cli.py run capacity`) finds the quantum reservoir **not more** — slightly ***less*** — nonlinearly expressive than a matched RFF/ESN at equal feature count: no excess expressivity to exploit. A frontier check (`cli.py run frontier`) adds that $g(n)$ does **not** widen toward n=16 (it declines; D_{eff} /rank grow but a matched ESN keeps pace), and a direct **resource-efficiency** frontier (quality vs feature count, inputs held fixed; `results/efficiency_frontier_findings.md`) shows a ***smaller*** QRC cannot substitute for a ***larger*** classical reservoir — quantum accuracy **saturates** while the classical curves keep improving (the two static maps are comparable per feature). The simulable-scale evidence shows no gap opening with scale.

6. Quantum platform and resource planning

Simulator-first on qBraid: dense statevector ≤12 qubits, sparse/TN to ≈16, GPU for larger.

Resource estimates (gate-Trotter, 20 layers; readout is single-basis — all $Z_i, Z_i Z_j$ are computational-basis-diagonal, so one S-shot set yields **all** $n+C(n,2)$ observables):

n	two-qubit gates	one-qubit gates	observables	shots S ($\epsilon \approx 1/\sqrt{S}$)	sim wall-clock
8	220	160	36	2k–8k ($\epsilon \approx .011\text{--}.022$)	<1 s/state
10	480	200	55	2k–8k	~sec
12	760	240	78	2k–8k	~1 min build
16 (sparse)	—	—	136	—	~40 min/point

(wall-clock is hardware-dependent — an engineering, not a scientific, quantity.)

QPU validation uses this gate-Trotter circuit on **IonQ / IQM / IBM** via qBraid — the random-sparse Ising needs ***fewer*** two-qubit gates than an all-to-all reservoir, easing NISQ mapping — with **ZNE + measurement mitigation** and a classical cross-check per run. The submission path is **executable now** (`qbraid_submit.py, cli.py run qsubmit`): on a simulator it (i) reproduces

the engine to 3.9×10^4 via the exact circuit (our per-run classical cross-check),
(ii) characterizes the **shot budget** (mean feature error 0.046/0.013/0.0066
at $S=256/4k/16k$; $\epsilon \approx 1/\sqrt{S}$; gate-Trotter(20) adds ≈ 0.04), and (iii) demonstrates **zero-noise extrapolation** recovering toward the noiseless value under a depolarizing sweep. It is one flag (- -device) from a real backend,
pending qBraid credit allocation. We thus characterize all four challenge axes: reservoir size, encoding density, shot budget, and noise. *(QCi Dirac-3 is the separate optimization challenge's device.)*

7. Limitations (stated plainly)

(i) **No quantum advantage** is demonstrated at the ≤ 16 -qubit simulable scale; HAR-X (classical, linear) is the best model on this task. (ii) The S&P 500 RV sample ends Feb-2020 — the 2008 GFC is in-sample, the 2020 COVID shock just outside it; broader assets/periods untested. (iii) Noise is studied on MNIST with single-qubit channels at the readout; full noisy-circuit and shot-noise simulation is deferred to the QPU runs. (iv) g is regularization-dependent (we report the qualitative gap, not a tuned value). (v) **No real-QPU run yet** (simulator cross-checked; pending qBraid credit allocation). (vi) Distinctness and full-rank entanglement are **necessary, not sufficient** for advantage — whether they convert beyond the classical-simulation frontier is the open question; a quantum-data probe (results/quantum_data_outlook_findings.md) marks where an edge most plausibly lies — the QRC natively reads nonlinear functionals of quantum **states** (purity, entanglement) beyond linear classical measurement, an advantage that should grow with input size (tomography costs 4^n) on hardware-native quantum data, the regime this challenge's classical data never probes.

8. Stakeholder impact, milestone plan, AI disclosure

Reliable volatility forecasts feed portfolio hedging, dynamic risk limits and derivatives pricing. We make the impact **concrete** (cli.py run economics): a vol-timing backtest sizing S&P-500 exposure by the one-step RV forecast **nearly halves the 2008 max drawdown (-61% → -32%)** at a 10% vol target, but a **plain HAR** forecast captures it — rich features and the quantum reservoir add **no** economic value (negative certainty-equivalent fees). The decision-useful lever is **vol-timing on a simple RV forecast, not quantum hardware. Milestone plan:** (i) pre-register ✓; (ii) scaling + encoding-density sweeps ✓; (iii) adversarial HAR-X/ESN/RFF test ✓; (iv) MNIST + noise ✓; (v) sparse/TN frontier + bond dimension ✓; (vi) gate-Trotter QPU validation (IonQ/IQM/IBM) — simulator cross-checked; fallback = TN + density-matrix noise emulation. **AI disclosure:** Claude (Anthropic) assisted with code and drafting under the team's direction; all formulations, decisions, and results are the team's own.

References

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